

PSP6 • ENTREGA 1 • BANCA

# Louças sanitárias em trânsito

Por que a avaria na entrega é um problema de projeto

**Avarias no transporte entre distribuidor e obra**



# O elo crítico foi delimitado entre o distribuidor e a obra.

DO FLUXO AO CONTATO



**Recorte:** danos ocultos na entrega, percebidos na descarga ou conferência. **Referências:** 1–3% de avaria geral em cargas bem embaladas; 2–5% em materiais de construção; e compra adicional de 3–5% recomendada para louças sanitárias. Não há taxa brasileira pública para distribuidor → obra.

# Escala logística e acidentes mostram por que a exposição importa

[S] FONTE SETORIAL  
[C] REFERÊNCIA COMERCIAL

|                                                                                           |                                                                                                               |                                                                                                                       |                                                                                                                                          |
|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| <div>[S] BRASIL</div> <div>22 mi</div> <div>de louças sanitárias produzidas por ano</div> | <div>[S] CONSTRUÇÃO</div> <div>18,9%</div> <div>dos acidentes do setor são cortes, lacerações e feridas</div> | <div>[S] TRABALHO</div> <div>51 s</div> <div>é o intervalo médio estimado entre acidentes de trabalho no Brasil</div> | <div>[C] BUFFER</div> <div>3–5%</div> <div>de compra extra recomendado para tiles e lavatórios cerâmicos diante do risco de quebra</div> |
|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|

Os dados de acidentes dimensionam o contexto de exposição do trabalhador; não atribuem essas ocorrências especificamente a louças sanitárias.

Fontes: ANFACER; AEAT/INSS, ano-base 2022 (apud Migalhas, 2024); MPT/Smartlab; HSY Sourcing.



# A avaria transfere custo, prazo e retrabalho para toda a entrega

## EVIDÊNCIA LOCAL

### Chega trincada

Lucas relatou peças já trincadas e quedas na descarga. Ele não presenciou acidentes nem quantificou a frequência.

## QUEM RESPONDE

### Distribuidor

Na visita de Rodrigo, o distribuidor relatou assumir crédito, reposição e uma nova entrega quando a avaria é identificada.

## POR QUE PERSISTE

### Proteção x custo

Proteções mais robustas existem, mas precisam caber no preço e na rotina de uma operação com muitos produtos.

## PRÁTICA CORRENTE

## Papelão e calços

Embalagem de baixo custo usada para separar, apoiar e reduzir o contato direto da louça.

**Limite: desempenho local não medido**

## RELATO DO DISTRIBUIDOR

## Proteção inflável

Foi apontada como tecnicamente eficaz para reduzir movimento e impacto durante o transporte.

**Limite: custo considerado inviável**

## ANTERIORIDADE

## Encaixes estruturais

Patentes mostram insertos moldados e estruturas corrugadas para imobilizar e amortecer.

**Limite: aderência ao Brasil não validada**

US20150014208A1 · referência de proteção

[illegible]

表41 1990年の若狭の漁獲の季節別

[illegible]

**JP2007186235A · referência de proteção**

(19) 中华人民共和国国家知识产权局

(12) 发明专利申请

(10) 中国公布号: CN 111094855  
(43) 中国公布日: 2022. 03. 18

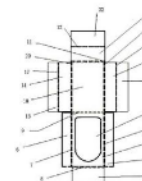
(21) 申请号: 20211011085. 3  
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(71) 申请人: 中国包机网测试中心  
地址: 300080 天津市滨海新区塘沽经济技术  
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董磊  
(74) 专利代理机构: 天津通理知识产权代理有限公司  
专利号: 32209  
代理人: 董磊  
(51) Int. Cl.  
B60D 5/02 (2006. 01)  
B60D 5/42 (2006. 01)

本申请涉及一种... 权利要求书 说明书 附图

(54) 发明名称  
一种马钱子碱用包膜结构及其制备方法

5. 实施例

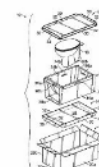
本发明涉及一种与桶形运输用包装结构, 包括板、背板、侧板、左立板、右立板、左底板及右板, 所述左板、背板、侧板或/和下层上均设置所述底板两侧可拆卸设置有左底板及右底板, 所述板与侧板均设置有相对立板及右立板。本发明涉及一种与桶形运输用包装结构的制备方法, 包括以下步骤: 1) 裁料; 2) 压板成型; 3) 背板; 4) 侧板; 5) 桶形运输用包装结构的制作完成。本发明设计科学合理, 工艺简单, 节约制作材料及成本, 并形美大方, 且操作简单, 经与桶形运输安全。



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| (15) <b>United States Patent</b><br>Gruff et al.                                | (36) <b>Patent No.:</b> <b>US 10,109,997 B2</b><br>(42) <b>Date of Patent:</b> <b>May 28, 2019</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| (24) <b>SHIPPING CONTAINER FOR PACKAGING<br/>INTERIOR PRODUCTS</b>              | C03C11/00, B65D 65/00 (2013.01); B65D<br>25/04 (2006.01); B65D 25/87 (2006.01)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| (71) <b>Applicant:</b> <b>INTERNATIONAL PAPER<br/>COMPANY, Memphis, TN (US)</b> | Field of Classification Search: <b>358/100, 358/101, 358/102, 358/103, 358/104, 358/105, 358/106, 358/107, 358/108, 358/109, 358/110, 358/111, 358/112, 358/113, 358/114, 358/115, 358/116, 358/117, 358/118, 358/119, 358/120, 358/121, 358/122, 358/123, 358/124, 358/125, 358/126, 358/127, 358/128, 358/129, 358/130, 358/131, 358/132, 358/133, 358/134, 358/135, 358/136, 358/137, 358/138, 358/139, 358/140, 358/141, 358/142, 358/143, 358/144, 358/145, 358/146, 358/147, 358/148, 358/149, 358/150, 358/151, 358/152, 358/153, 358/154, 358/155, 358/156, 358/157, 358/158, 358/159, 358/160, 358/161, 358/162, 358/163, 358/164, 358/165, 358/166, 358/167, 358/168, 358/169, 358/170, 358/171, 358/172, 358/173, 358/174, 358/175, 358/176, 358/177, 358/178, 358/179, 358/180, 358/181, 358/182, 358/183, 358/184, 358/185, 358/186, 358/187, 358/188, 358/189, 358/190, 358/191, 358/192, 358/193, 358/194, 358/195, 358/196, 358/197, 358/198, 358/199, 358/200, 358/201, 358/202, 358/203, 358/204, 358/205, 358/206, 358/207, 358/208, 358/209, 358/210, 358/211, 358/212, 358/213, 358/214, 358/215, 358/216, 358/217, 358/218, 358/219, 358/220, 358/221, 358/222, 358/223, 358/224, 358/225, 358/226, 358/227, 358/228, 358/229, 358/230, 358/231, 358/232, 358/233, 358/234, 358/235, 358/236, 358/237, 358/238, 358/239, 358/240, 358/241, 358/242, 358/243, 358/244, 358/245, 358/246, 358/247, 358/248, 358/249, 358/250, 358/251, 358/252, 358/253, 358/254, 358/255, 358/256, 358/257, 358/258, 358/259, 358/260, 358/261, 358/262, 358/263, 358/264, 358/265, 358/266, 358/267, 358/268, 358/269, 358/270, 358/271, 358/272, 358/273, 358/274, 358/275, 358/276, 358/277, 358/278, 358/279, 358/280, 358/281, 358/282, 358/283, 358/284, 358/285, 358/286, 358/287, 358/288, 358/289, 358/290, 358/291, 358/292, 358/293, 358/294, 358/295, 358/296, 358/297, 358/298, 358/299, 358/300, 358/301, 358/302, 358/303, 358/304, 358/305, 358/306, 358/307, 358/308, 358/309, 358/310, 358/311, 358/312, 358/313, 358/314, 358/315, 358/316, 358/317, 358/318, 358/319, 358/320, 358/321, 358/322, 358/323, 358/324, 358/325, 358/326, 358/327, 358/328, 358/329, 358/330, 358/331, 358/332, 358/333, 358/334, 358/335, 358/336, 358/337, 358/338, 358/339, 358/340, 358/341, 358/342, 358/343, 358/344, 358/345, 358/346, 358/347, 358/348, 358/349, 358/350, 358/351, 358/352, 358/353, 358/354, 358/355, 358/356, 358/357, 358/358, 358/359, 358/360, 358/361, 358/362, 358/363, 358/364, 358/365, 358/366, 358/367, 358/368, 358/369, 358/370, 358/371, 358/372, 358/373, 358/374, 358/375, 358/376, 358/377, 358/378, 358/379, 358/380, 358/381, 358/382, 358/383, 358/384, 358/385, 358/386, 358/387, 358/388, 358/389, 358/390, 358/391, 358/392, 358/393, 358/394, 358/395, 358/396, 358/397, 358/398, 358/399, 358/400, 358/401, 358/402, 358/403, 358/404, 358/405, 358/406, 358/407, 358/408, 358/409, 358/410, 358/411, 358/412, 358/413, 358/414, 358/415, 358/416, 358/417, 358/418, 358/419, 358/420, 358/421, 358/422, 358/423, 358/424, 358/425, 358/426, 358/427, 358/428, 358/429, 358/430, 358/431, 358/432, 358/433, 358/434, 358/435, 358/436, 358/437, 358/438, 358/439, 358/440, 358/441, 358/442, 358/443, 358/444, 358/445, 358/446, 358/447, 358/448, 358/449, 358/450, 358/451, 358/452, 358/453, 358/454, 358/455, 358/456, 358/457, 358/458, 358/459, 358/460, 358/461, 358/462, 358/463, 358/464, 358/465, 358/466, 358/467, 358/468, 358/469, 358/470, 358/471, 358/472, 358/473, 358/474, 358/475, 358/476, 358/477, 358/478, 358/479, 358/480, 358/481, 358/482, 358/483, 358/484, 358/485, 358/486, 358/487, 358/488, 358/489, 358/490, 358/491, 358/492, 358/493, 358/494, 358/495, 358/496, 358/497, 358/498, 358/499, 358/500, 358/501, 358/502, 358/503, 358/504, 358/505, 358/506, 358/507, 358/508, 358/509, 358/510, 358/51</b> |

15 Claims, 8 Drawing Sheet



US10301097B2 · referência de proteção

CONCLUSÃO DA DESCOBERTA

# Problema definido

OPPORTUNITY STATEMENT

**“Como podemos  
reduzir as avarias de  
louças sanitárias no  
transporte entre o  
distribuidor e a obra,  
sem elevar o custo da  
operação e reduzindo o  
manuseio de peças  
danificadas?”**

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